



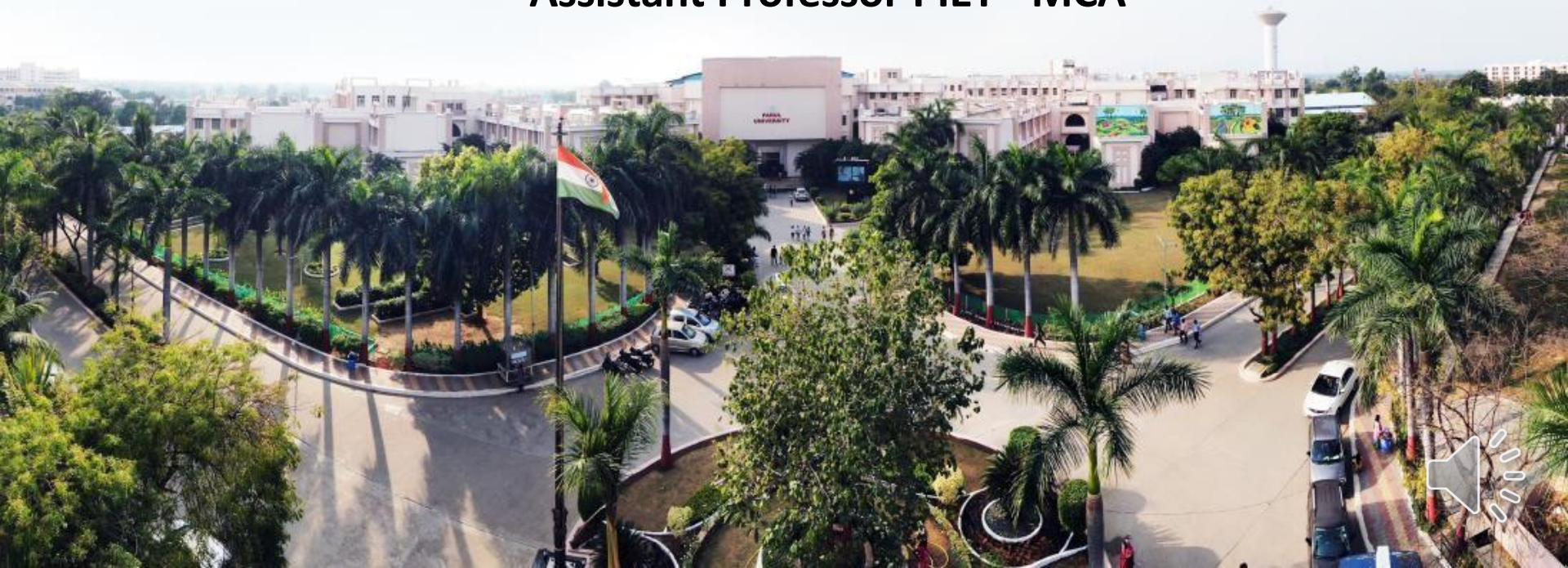
05201302

# Web Application Development

MCA MSc-IT Semester 3

**Prof Faruk Abdulla**

**Assistant Professor PIET - MCA**





## Introduction

### **What You Will Learn:**

Asynchronous interaction with AJAX

Powerful querying with LINQ

Efficient database access with Stored Procedures

Managing XML data effectively

Preventing spam attacks using Captcha

Organizing applications using ASP.NET MVC



## Introduction

ASP.NET helps build robust, interactive web applications. This tutorial covers key advanced techniques to improve app performance, security, and architecture for modern development.



## ASP.NET AJAX Extension

AJAX allows webpages to update data asynchronously without full reloads, improving speed and smooth experience.

- Makes pages feel dynamic and responsive
- Reduces server load by limiting updates
- Used widely in modern web apps like Facebook and Gmail



Example Scenario: When clicking a button, only a data grid should refresh without reloading the whole page.

Steps:

- Add a ScriptManager control to your ASPX page.
- Wrap the controls to be updated inside an UpdatePanel.
- Add a button inside the UpdatePanel.





```
<asp:ScriptManager ID="ScriptManager1" runat="server" />
<asp:UpdatePanel ID="UpdatePanel1" runat="server">
  <ContentTemplate>
    <asp:Label ID="lblTime" runat="server" Text="Time will update
here"></asp:Label>
    <asp:Button ID="btnRefresh" runat="server" Text="Refresh Time"
OnClick="btnRefresh_Click" />
  </ContentTemplate>
</asp:UpdatePanel>
```

```
protected void btnRefresh_Click(object sender, EventArgs e)
{
    lblTime.Text = DateTime.Now.ToString("T");
}
```

Result: Only label updates asynchronously when button clicked, no full-page refresh.

## LINQ (Language Integrated Query)

LINQ integrates database-like queries into C#, making data retrieval easy and error-free.

- Simplifies coding, avoids manual loops
- Works with in-memory collections, XML, or databases

## LINQ (Language Integrated Query)

LINQ integrates database-like queries into C#, making data retrieval easy and error-free.

- Simplifies coding, avoids manual loops
- Works with in-memory collections, XML, or databases



## Stored Procedures in ASP.NET

Predefined SQL commands stored in the database to improve security and performance. ASP.NET calls these instead of writing raw SQL in code.

- Prevent SQL Injection
- Optimize repeated queries
- Centralize business logic at DB level

## Stored Procedures in ASP.NET

Predefined SQL commands stored in the database to improve security and performance. ASP.NET calls these instead of writing raw SQL in code.

- Prevent SQL Injection
- Optimize repeated queries
- Centralize business logic at DB level

## Stored Procedures in ASP.NET

Scenario: Retrieve student records by student ID using stored procedure.

```
CREATE PROCEDURE GetStudentById  
    @StudentID INT  
AS  
BEGIN  
    SELECT * FROM Students WHERE ID = @StudentID  
END
```



## Stored Procedures in ASP.NET

```
using(SqlConnection conn = new SqlConnection(connString))
{
    SqlCommand cmd = new SqlCommand("GetStudentById", conn);
    cmd.CommandType = CommandType.StoredProcedure;
    cmd.Parameters.AddWithValue("@StudentID", 2);

    conn.Open();
    SqlDataReader reader = cmd.ExecuteReader();
    if(reader.Read())
    {
        Console.WriteLine(reader["Name"].ToString());
    }
}
```

# XML

XML helps share data across different systems due to its readability and standardized format.

Easy to read/write

Supported in almost all languages and platforms

# XML

```
<students>
  <student>
    <name>Alice</name>
    <age>19</age>
  </student>
  <student>
    <name>Bob</name>
    <age>22</age>
  </student>
</students>
```





## XML

```
XDocument doc = XDocument.Load("students.xml");  
var names = from s in doc.Descendants("student")  
            select s.Element("name").Value;  
  
foreach(var name in names)  
{  
    Console.WriteLine(name);  
}
```

## Captcha

CAPTCHA ensures form submissions come from humans, not automated bots.

Prevents spam

Protects registration/login forms

## Captcha

### Integrating Google reCAPTCHA

- Steps:
- Register your site on Google reCAPTCHA to get site and secret keys.
- Include the reCAPTCHA widget on your ASPX page.
- Validate user response on form submission.



## Captcha

### Integrating Google reCAPTCHA

```
<script  
src="https://www.google.com/recaptcha/api.js"></script>  
<form id="form1" runat="server">  
    <div class="g-recaptcha" data-sitekey="your-site-key"></div>  
    <asp:Button runat="server" Text="Submit"  
OnClick="Submit_Click" />  
</form>
```



## Captcha

```
protected void Submit_Click(object sender, EventArgs e)
{
    string response = Request["g-recaptcha-response"];
    bool isValid = ValidateCaptcha(response); // Implement server-side call to Google
API
    if(isValid)
    {
        // Process form
    }
    else
    {
        // Show error
    }
}
```

# ASP.NET MVC Framework

Separates concerns into:

Model: Manages data and business logic

View: Displays UI and data to the user

Controller: Handles user requests and updates model or view

## ASP.NET MVC Framework

```
public class StudentController : Controller
{
    public ActionResult List()
    {
        var students = new List<string> { "Alice", "Bob", "Charlie" };
        return View(students);
    }
}
```



# ASP.NET MVC Framework

## MVC Routing

URL pattern maps to controller/action/id

Example URL: /Student/List/2

Controller: StudentController

Action: List

Id: Optional parameter (e.g., student id)